PEAS Framework Analysis of AI Agents

Performance measure, Environment, Actuators, Sensors — plus environment-type classification for each agent.

1. Email Spam Filter

Performance Measure	Correctly classifying spam vs. legitimate emails; minimizing false positives (blocking real emails) and false negatives (letting spam through)
Environment	Email inbox/server, incoming email stream, user feedback (marking as spam/not spam)
Actuators	Move email to spam folder, flag/label email, delete email
Sensors	Email content (text, sender, links, attachments), metadata (headers, timestamps), user actions (mark as spam)

Environment Types: Partially Observable • Stochastic • Episodic (classifying one email doesn't affect the next) • Discrete • Single-agent • Static (email doesn't change while being analyzed)

2. Smart Home Climate Control System

Performance Measure	Maintaining desired temperature/humidity, energy efficiency, occupant comfort
Environment	House/room, outdoor weather, occupants' presence and preferences
Actuators	Turn on/off heater/AC, adjust fan speed, open/close vents
Sensors	Temperature sensors, humidity sensors, occupancy/motion sensors, weather data feed

Environment Types: Partially Observable • Stochastic • Sequential (current temperature adjustment affects future states) • Continuous (temperature values) • Single-agent • Dynamic (temperature changes even if the agent is "thinking")

3. Stock Trading Bot

Performance Measure	Maximizing profit/return, minimizing risk/loss, portfolio growth
Environment	Stock market, price feeds, news, other traders/bots
Actuators	Buy, sell, hold orders
Sensors	Real-time stock prices, market news, historical data, trading volume

Environment Types: Partially Observable • Stochastic • Sequential (each trade affects portfolio and future decisions) • Continuous (prices) • Multi-agent (competing with other traders/bots) • Dynamic (market moves independently of the agent's decisions) • Unknown (agent often doesn't fully know market "rules"/other players' strategies)

4. Autonomous Content Recommendation Engine

Performance Measure	User engagement (clicks, watch time), satisfaction, retention

Environment	Platform's content library, user profile/history, other users' behavior
Actuators	Display/recommend specific content, rank/order suggestions
Sensors	User clicks, watch/read time, likes/dislikes, browsing history

Environment Types: Partially Observable • Stochastic • Sequential (recommendations shape future user behavior/preferences) • Discrete (finite content items) • Multi-agent (many users interact with the shared system) • Dynamic (content library and trends evolve over time)

5. Mars Rover

Performance Measure	Successful navigation, data/sample collection, avoiding damage, mission objectives completed
Environment	Martian terrain, rocks, slopes, dust storms, limited communication with Earth
Actuators	Wheels/motors (movement), robotic arm, drilling tools, cameras (positioning)
Sensors	Cameras, LIDAR, spectrometers, temperature/pressure sensors, GPS-like positioning

Environment Types: Partially Observable • Stochastic (terrain/weather unpredictability) • Sequential (movement decisions affect future navigation options) • Continuous (terrain, movement) • Single-agent • Dynamic (weather/dust storms change independent of rover's decisions) • Unknown (rover often exploring unfamiliar terrain)

6. Online Shopping Recommender

Performance Measure	Purchase conversion rate, click-through rate, customer satisfaction, revenue
Environment	E-commerce platform, product catalog, customer browsing/purchase history
Actuators	Display recommended products, personalize homepage/search results
Sensors	Click history, purchase history, search queries, cart activity, ratings/reviews

Environment Types: Partially Observable • Stochastic • Sequential (a recommendation now can influence future browsing/purchase behavior) • Discrete (finite product catalog) • Multi-agent (many customers interacting with the same system) • Dynamic (inventory, prices, trends change over time)

Quick Comparison Table

Agent	Observable	Det./Stoch.	Episodic/Sequential	Discrete/Cont.	Agents	Static/Dynamic
Spam Filter	Partial	Stochastic	Episodic	Discrete	Single	Static
Climate Control	Partial	Stochastic	Sequential	Continuous	Single	Dynamic
Stock Trading Bot	Partial	Stochastic	Sequential	Continuous	Multi	Dynamic
Content Recommender	Partial	Stochastic	Sequential	Discrete	Multi	Dynamic
Mars Rover	Partial	Stochastic	Sequential	Continuous	Single	Dynamic
Shopping Recommender	Partial	Stochastic	Sequential	Discrete	Multi	Dynamic

Note on the Spam Filter being "Episodic": This follows the same reasoning applied to the vacuum cleaner example — classifying one email doesn't require memory of the previous email's classification decision. A stricter interpretation could argue **Sequential** instead, since spam patterns/history could inform future filtering — worth mentioning both angles if it comes up in an exam.